

Installation of Splunk SIEM and Log Forwarding

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Introduction to the Lab

Splunk is an innovative technology which searches and indexes log files and helps organizations derive insights from the data. A main benefit of Splunk is that it uses indexes to store data, and so does not require a separate database to store its information. Splunk is used for monitoring and searching through big data. It indexes and correlates information in a container that makes it searchable, and makes it possible to generate alerts, reports, and visualizations. It can recognize data patterns, create metrics, and help diagnose problems, for business challenges like IT management, security, and compliance.

Splunk forwarders consume data and send it to an indexer. Forwarders require minimal resources and have little impact on performance, so they can usually reside on the machines where the data originates. There are two kinds of forwarders in Splunk.

- i) The **universal forwarder** contains only the components that are necessary to forward data. The universal forwarder is designed to run on production servers, having minimal CPU and memory usage and the least impact possible on mission-critical software.
- ii) A **heavy forwarder is a full Splunk Enterprise instance** that can index, search, and change data as well as forward it. Splunk forwarder in our Splunk server ubuntu machine acts as an agent for log collection from

remote machines. Splunk forwarder collects logs from remote machines and forwards them to the indexer (Splunk database) for further processing and storage.

Note: Universal forwarders do not have a web or application interface. Once installed, you must make configuration changes at the command line in both Windows and Unix- or Linux-based systems.

Best practices:

- Use the universal forwarder when possible as a data collection method.
- Stop and start the universal forwarder from the command line.

In this lab, you will install Splunk SIEM as a standalone machine on a Linux Ubuntu OS. You will be configuring Universal Forwarder on another Ubuntu machine for sending authentication logs to Splunk server. These logs would be captured in Splunk server.

Pre-requisites of the lab

The pre-requisites (recommended system requirement) for configuring Universal Forwarder and Splunk Server for log collection are:

- VMware Fusion or Workstation (It must be registered; trials will not work)
- Install two Ubuntu Virtual machines
- RAM 2+GB
 - 2 GHz dual core processor or higher
 - 4 GB system memory
 - 20 GB of free hard drive space
 - Internet access
- Make sure both virtual machines are on the same network, Preferably NAT

Steps of the lab:

To get our VM up and running, we will need to execute the following steps:

Step 1: Preparing the first Ubuntu VM for Splunk Installation

Step 2: Configuring receiving port at Ubuntu Splunk Server

Step 3: Download and install Splunk Universal Forwarder on second ubuntu machine

Step 4: Searching and Reporting App – Capture the logs in Splunk Ubuntu Server

Execution of the lab (Saving your screenshots here)

Step 1: Preparing the first Ubuntu VM for Splunk Installation

Open the command terminal and run the following commands

a) Change to the Downloads directory using the command `cd Downloads/`

b) Run the command below to download Splunk.

```
wget -O splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

```
'https://download.splunk.com/products/splunk/releases/8.2.4/linux/splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb'
```

c) Run the command below to install the downloaded Splunk package

```
sudo apt install ./splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

[Attach your Screenshot]

```
ubuntu@ubuntu:~/Desktop$ wget -O splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/9.4.0/linux/splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb"
--2025-02-10 17:16:30-- https://download.splunk.com/products/splunk/releases/9.4.0/linux/splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 18.245.96.61, 18.245.96.39, 18.245.96.79, ...
Connecting to download.splunk.com (download.splunk.com)|18.245.96.61|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 920120936 (877M) [binary/octet-stream]
Saving to: 'splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb'

splunk-9.4.0-6b4ebe 100%[=====] 877.50M 46.7MB/s in 18s

2025-02-10 17:16:48 (48.3 MB/s) - 'splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb' saved [920120936/920120936]

ubuntu@ubuntu:~/Desktop$ sudo apt install ./^C
ubuntu@ubuntu:~/Desktop$ sudo apt install ./splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'splunk' instead of './splunk-9.4.0-6b4ebe426ca6-linux-amd64.deb'
The following NEW packages will be installed:
  splunk
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/920 MB of archives.
After this operation, 0 B of additional disk space will be used.
0% [Working]
```

- d) Switch to root using the command `sudo su` and input your password
- e) Switch to the Splunk directory using the command `cd /opt/splunk/bin`
- f) Run the command to accept Splunk license and
`./splunk start --accept-licence/`
- g) Press enter to get to the end of the License agreement and type y (yes) to accept the license
- h) Input an administrator username with the format `Firstname-NETW2200` and a password you can remember as you won't be able to access Splunk without this.

[Attach your Screenshot]

```
Please enter an administrator username: Madhvi-NETW2200
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (0x10001)
writing RSA key

Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (0x10001)
writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.

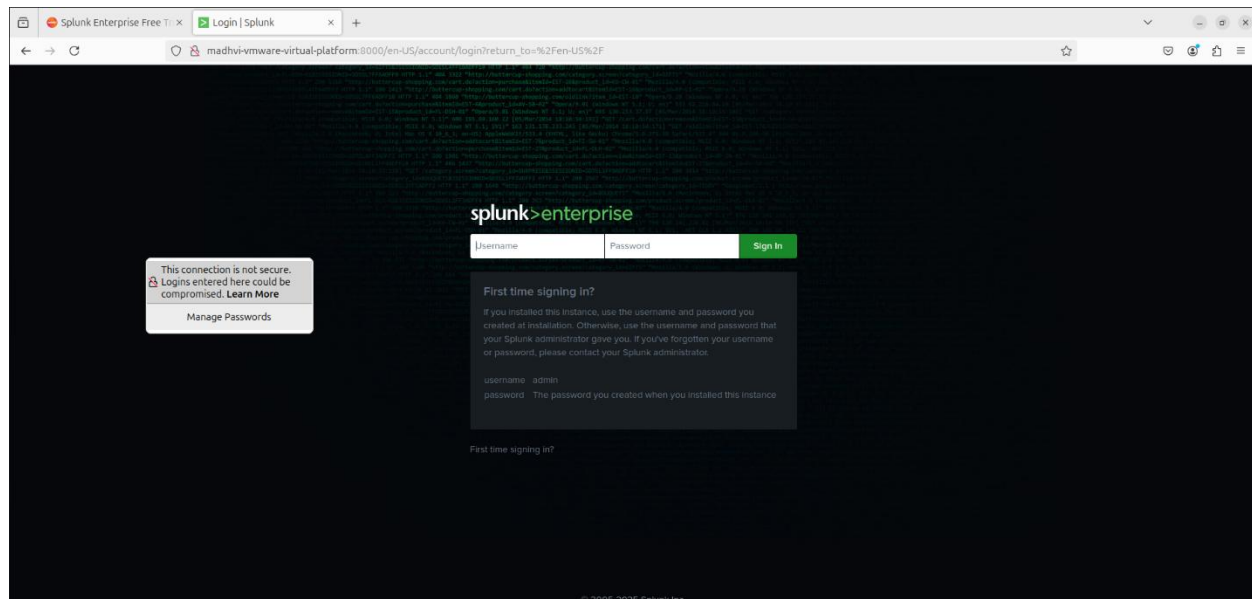
Splunk> 4TW
```

And wait for some time till it's done installing (Sometimes may work right away and some time may get errors)

- i) Splunk should be ready and the web url would be in your command line. Copy the web url and paste on your browser to access Splunk.

Alternatively, Splunk can also be accessed in the browser using the ubuntu Virtual machines IP address routed through port 8000 i.e., 192.168.80.130:8000.

[Attach your Screenshot]



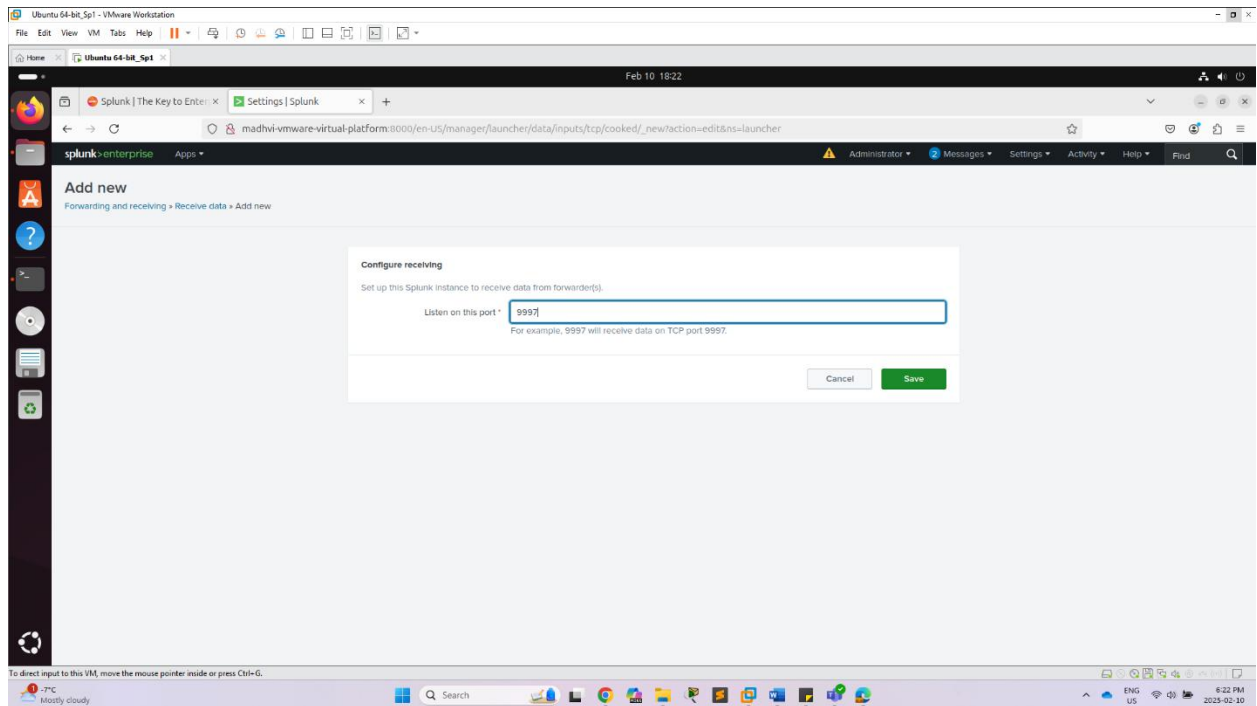
j) Go back to the command line and type the command below to ensure Splunk services starts at any reboot of our virtual machine `./splunk enable boot-start`

Step 2: Configuring receiving port at Ubuntu Splunk Server

- a) Go to web interface of Splunk enterprise by using `ipaddress:8000`. It will ask your administrator username and password for Splunk. Go to settings tab above and click on forwarding and receiving
- b) Click on Configure receiving under the Receive data column.
- c) Click on new receiving ports

- d) Input the default receiving port on TCP port 9997

[Attach your Screenshot]



- e) Go back to your command line and run the **command netstat -auntp | grep 9997**. This would allow us to see the new configured receiving port we added in previous step.

[Attach your Screenshot]

```
Setting up net-tools (2.10-0.1ubuntu4) ...
Processing triggers for man-db (2.12.0-4build2) ...
root@madhvi-VMware-Virtual-Platform:/opt/splunk/bin# netstat -auntp | grep 9997
tcp        0      0 0.0.0.0:9997          0.0.0.0:*            LISTEN     53216/splunkd
root@madhvi-VMware-Virtual-Platform:/opt/splunk/bin#
```

Now our Splunk is listening on the port 9997 for any logs from our universal forwarder. **Congratulations, you have just configured Splunk Ubuntu Server to receive logs now.**

That's the end of installing Splunk. Next, would be to install our Splunk universal forwarder to forward logs to Splunk.

Step 3: Download and install Splunk Universal Forwarder on second ubuntu machine

- a) Go to your second ubuntu machine you installed, open the terminal window and change to the downloads directory using the command `cd Downloads` to run the command (I believe you all should know by now what `cd` (change directory) command means)

```
wget -O splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

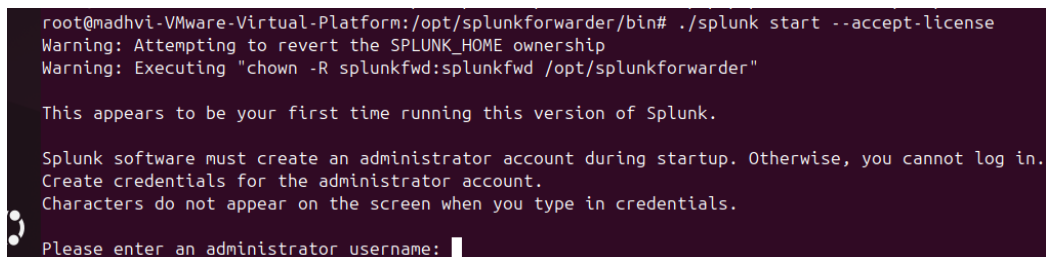
```
'https://download.splunk.com/products/universalforwarder/releases/8.2.4/linux/splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb'
```

- b) Run the command `sudo su` and input password to change to root user.
- c) Install the universal forwarder using the command.

```
apt install ./splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

- d) Run the command `cd /opt/splunkforwarder/bin/` to change to the Splunk forwarder bin directory and start the license using the command `./splunk start --accept-license` to accept the license agreement

[Attach your Screenshot]



```
root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: █
```


- e) set your forwarders username to `firstname-NETW2200` and input password

[Attach your Screenshot]

```

root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin
Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: Madhvi-NETW2200
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Important: splunk will start under systemd as user: splunkfwd
The unit file has been created.

Splunk> 4TW

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/jupload
    Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
    Creating: /opt/splunkforwarder/var/run/splunk/search_log
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dirmoncache
    Creating: /opt/splunkforwarder/var/lib/splunk/authDb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashDb
    Creating: /opt/splunkforwarder/var/run/splunk/collect
    Creating: /opt/splunkforwarder/var/run/splunk/sessions
  New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4.0-6b4ebe426ca6-linux-and64-manifest'
  All installed files intact.
    Done
  All preliminary checks passed
  
```

- f) Next step would be to add our Splunk's IP address to our forwarder to forward logs using the command `./splunk add forward-server (your splunk IP-address:port added to splunk to receive logs)`

e.g. `./splunk add forward-server 192.168.80.130:9997`

[Attach your Screenshot]

```

root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk add forward-server 192.168.81.137:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: Madhvi_NETW2200
Password:
Added forwarding to: 192.168.81.137:9997.
root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin#
  
```

g) Run the command to enable boot at start

```
/opt/splunkforwarder/bin/splunk enable boot-start
```

h) Next would be to run the command `./splunk start` on the universal forwarder to make sure we start the service.

i) Next step would be to run the command `./splunk add monitor`

```
/var/log/auth.log -sourcetype linux_secure
```

to send our authentication logs to Splunk (**Note:** This may ask for your Splunk username and password to authenticate)

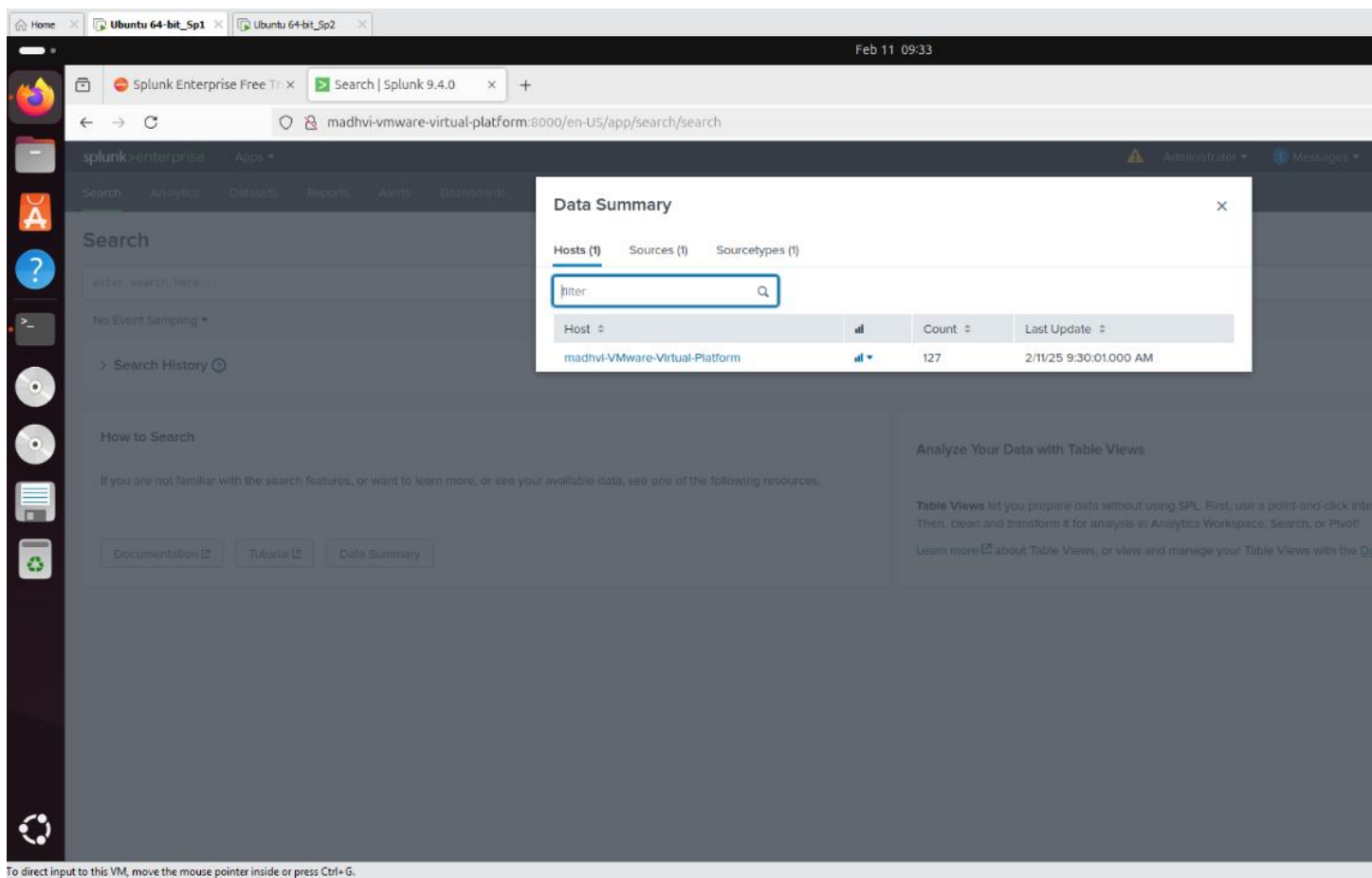
[Attach your Screenshot]

```
root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk add monitor /var/log/auth.log -sourcetype linux_secure
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/auth.log'.
root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin#
```

Step 4: Searching and Reporting App – Capture the logs in Splunk Ubuntu Server

- a) Once we run add-monitor command in previous, the next thing we would do is to wait for few minutes for this to take effect. After few minutes, click on Data summary on the search page of Ubuntu Splunk Server and you would see your authentication logs data has started ingesting into Splunk as seen below. Keep time picker as 24 hours.

[Attach your Screenshot]



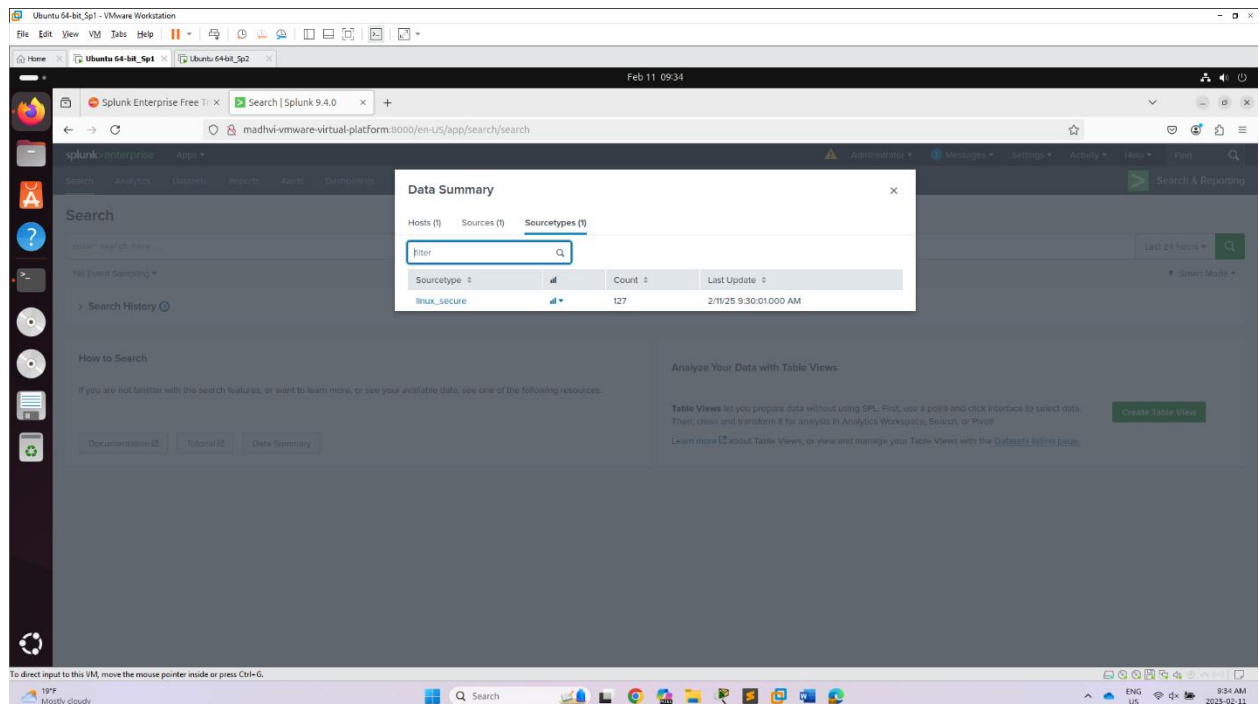
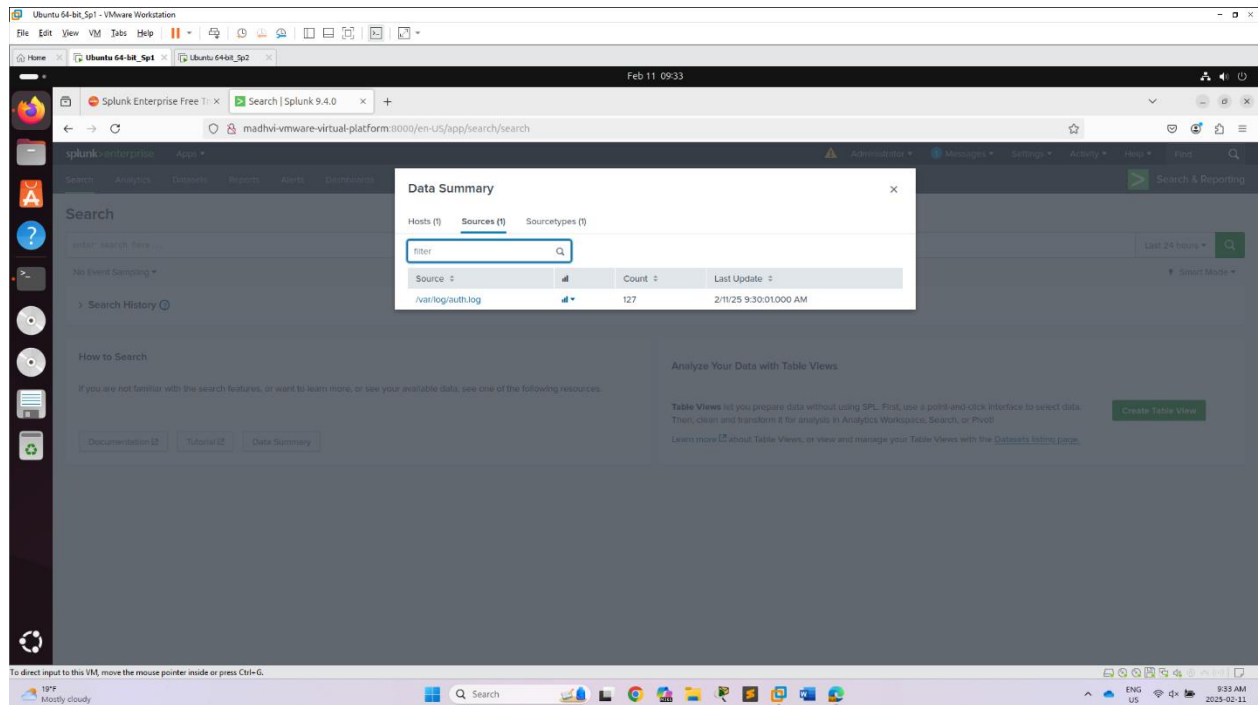
The screenshot shows the Splunk Search interface. The top navigation bar includes 'Search', 'Analytics', 'Dashboards', 'Reports', 'Alerts', and 'Dashboards'. The main content area displays the 'Data Summary' app. A search bar is visible with the text 'enter search here...'. Below the search bar, there is a 'Search History' section. The 'Data Summary' app shows a table with the following data:

Host	Count	Last Update
madhvi-VMware-Virtual-Platform	127	2/11/25 9:30:01.000 AM

Lab 3: Configuring Universal Forwarder and Splunk server for Log collection (Individual Lab)

NETW 2200 – Fuad Mustapha

Total Marks = 30



b) Now we run the command `index=*`, this way we would be able to see some data in our Splunk (To make the most of our free Splunk license in this lab,

fuad.mustapha@durhamcollege.ca

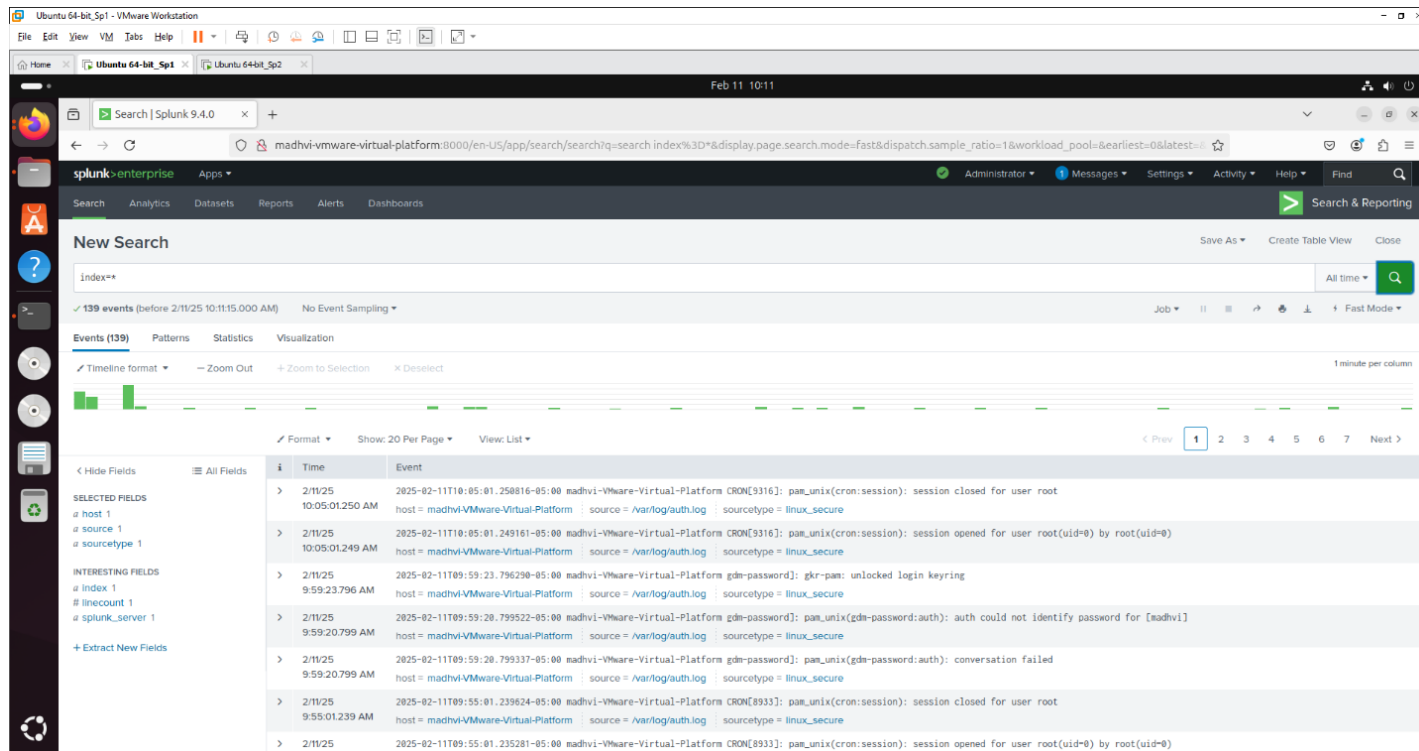
Lab 3: Configuring Universal Forwarder and Splunk server for Log collection (Individual Lab)

NETW 2200 – Fuad Mustapha

Total Marks = 30

we'll send only our authentication logs to Splunk for the time being, and we'll explore with more log forwarding in future Labs if we need more detections.)

[Attach your Screenshot]

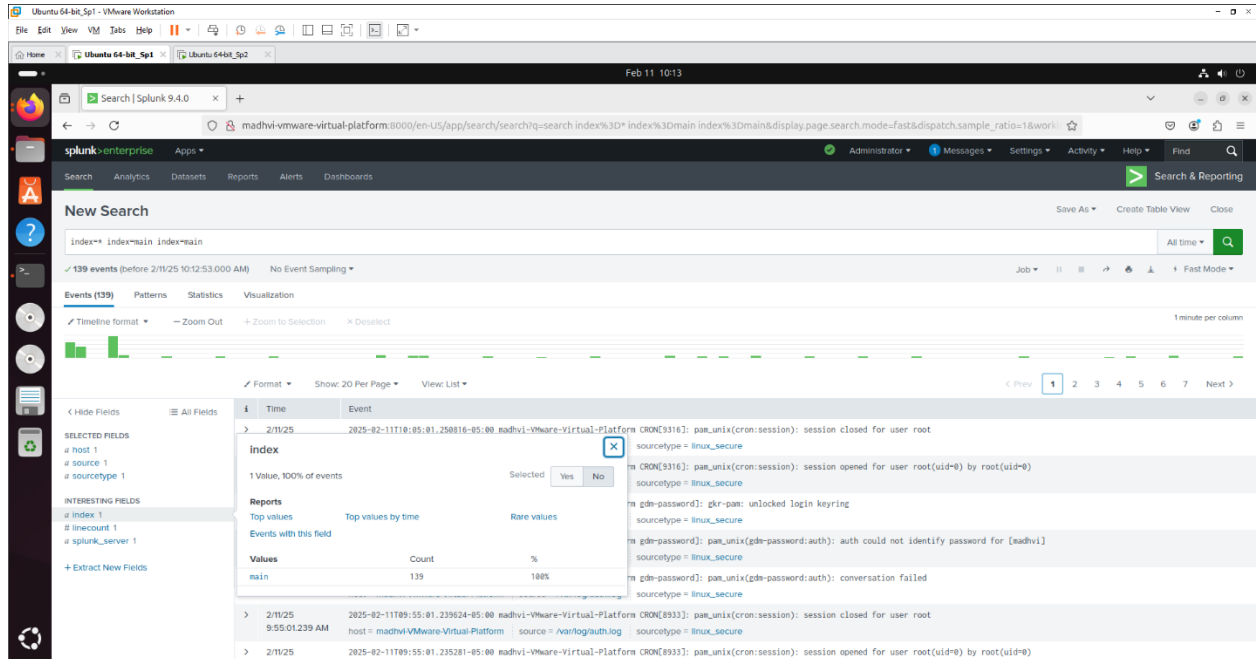


Congratulations, you've just set up Splunk and the Splunk universal forwarder to forward your authentication logs from Ubuntu Linux to the SIEM (Splunk)

Questions

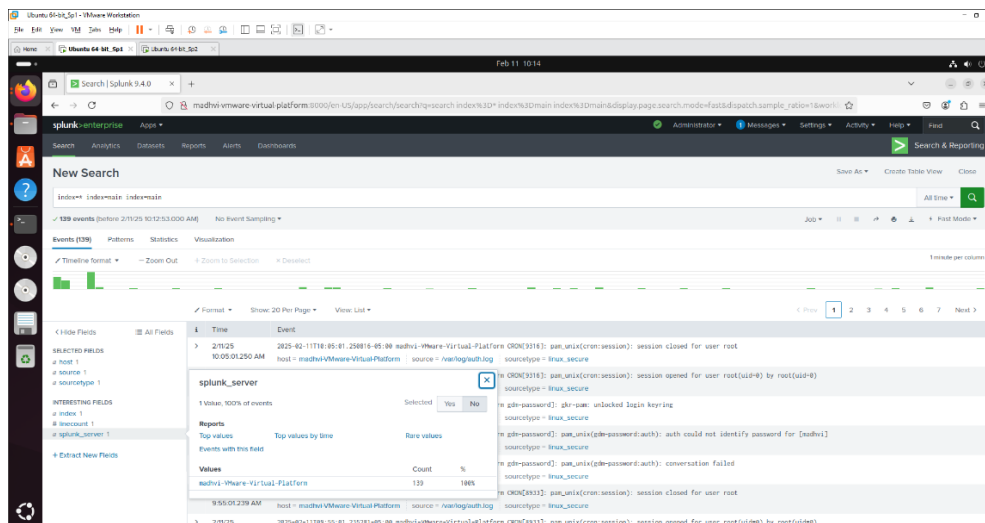
Answer the following questions using your search you ran above from the command `index=* search`.

- What is your Splunk index? (3marks)



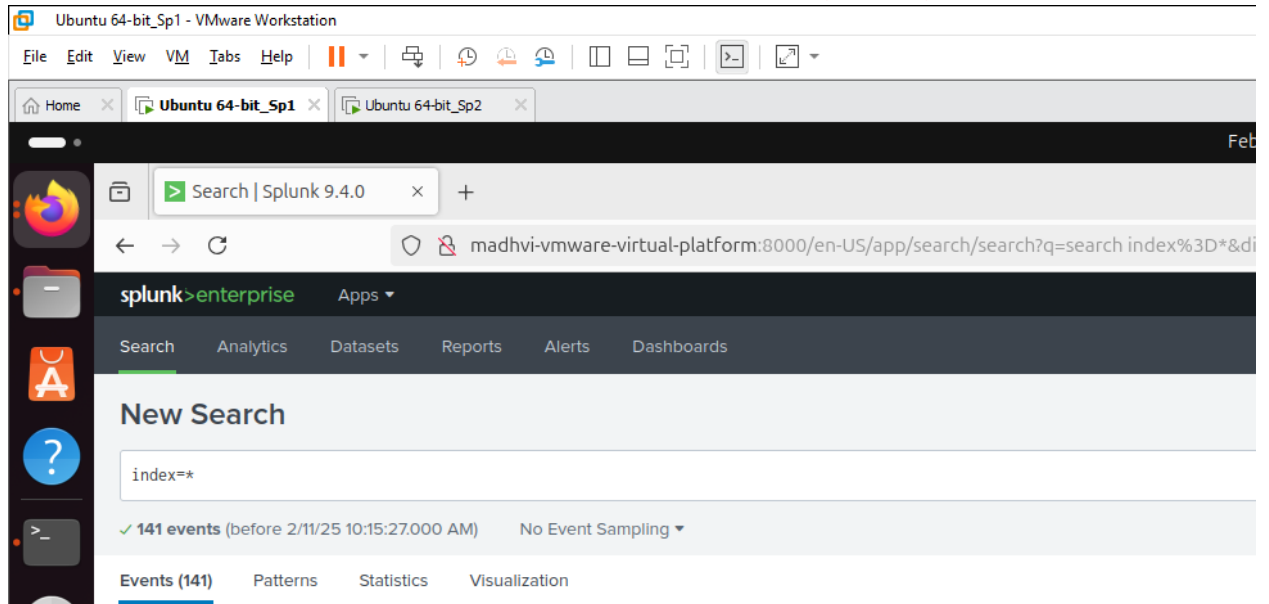
The screenshot shows the Splunk Enterprise search interface. The search bar contains the query `index=* index=main index=main`. The search results show 139 events. The 'Index' field is highlighted in the 'Selected' column of the 'Fields' table, indicating it is the selected index for the search.

- What is your splunk_server name? (1mark)



The screenshot shows the Splunk Enterprise search interface. The search bar contains the query `index=* index=main index=main`. The search results show 139 events. The 'splunk_server' field is highlighted in the 'Selected' column of the 'Fields' table, indicating it is the selected server for the search.

- What does the * in the index=* means? (Hint: it's one word) (1 mark)
* in index=* means "all" and tells Splunk to search across all indexes.
- What type of logs are we currently collecting in Splunk? (1 mark)
Our Splunk instance is collecting Linux authentication logs from /var/log/auth.log
- Based on your screenshot for the search (index=*), how many events can you see? (1mark)
Based on the screenshot, the number of events visible in the search (index=*) is 141 events



Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 16 Screenshots each 1 mark = 16 marks
 - b) Questions = 7 marks
 - c) Video Demonstration = 7 marks
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professors from lab manual. They are for your reference only and would be considered an Academic Integrity Breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**
- **Submit a video recording (sharing screen) briefly explaining the steps taken to perform the lab. If the video recording stating the above requirements are not submitted, you will be eligible to get marks out of 60% of Lab2 marks.**

*****GOOD LUCK*****

Student Name: madhvi Lashkaree

Student ID: 100997946

Introduction to the Lab

This lab is in continuation of Lab 3(Configuration of Splunk Universal Forwarder).

Part A: We will install openssh-server on our Splunk Forwarder Virtual Machine. After installing this, we would attempt logging in using ssh from a Kali machine using invalid and valid credentials. This Kali attack will be captured by our Splunk Ubuntu Server and would be captured as logs in it.

Pre-requisites of the lab

The pre-requisites (recommended system requirement) are:

- Lab 3 Virtual Machine.
- Kali Virtual Machine (Attacker machine) with following specifications:
 - 4 GHz dual core processor or better
 - 4 GB system memory
 - 80 GB of free hard drive space
 - Internet access is helpful

Steps of the lab:

To get our VM up and running, we will need to execute the following steps:

Step 1: Download a Kali Image and launch the VM.

Step 2: Install openssh-server on second Ubuntu Universal Forwarder machine.

Step 3: Login to Kali and Password Spray our Ubuntu Universal Forwarder machine (Victim Machine)- invalid Credentials (RED TEAM)

Step 4: Check the attack captured in first Ubuntu Splunk Server Machine.

Step 5: Brute force attack on second Ubuntu Universal Forwarder machine by Kali machine by using Hydra tool in Kali Linux- Successful attack

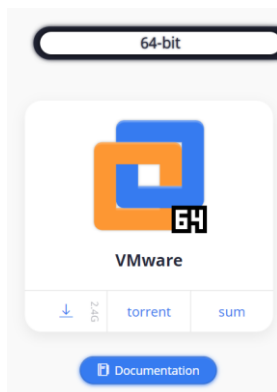
Step 6: Check the successful attack logs in your first Ubuntu Splunk server machine

Step 7: Answer few questions based on the scenario we saw.

Execution of the lab(Attach Your Screenshots)

Step 1: Download a Kali Image and launch the VM.

- a) Let us download a kali image from kali.org, Use the link below to download kali - <https://kali.download/virtual-images/kali-2021.4a/kali-linux-2021.4a-vmware-amd64.7z>

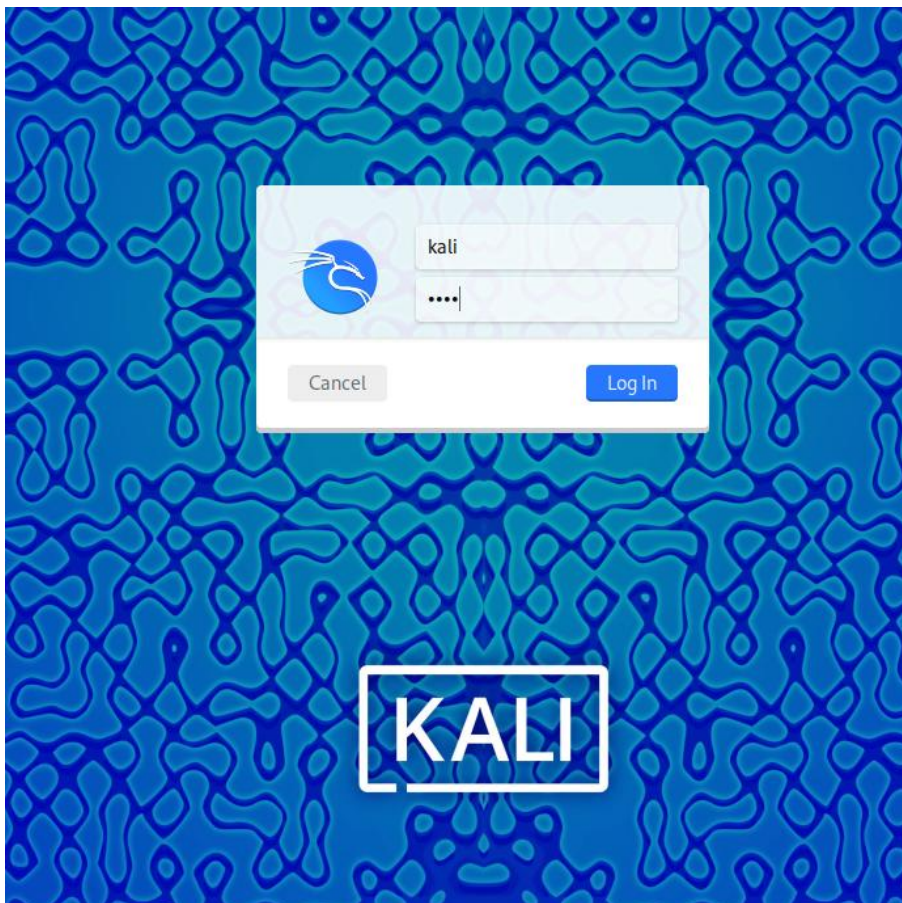


Note: If you have kali VM from other course, you can use that for this lab. In that case, please make sure it has the necessary tools and configurations required for the lab. It needs to be in the same network as our other two Ubuntu machines.

- b) Unzip your downloaded kali image using 7zip or any other unzipping tool. Go to VMware and click on file -> open. Navigate to your downloaded and unzipped kali to open it and your Kali VM would be created.

Note: This is a kali image downloaded from kali website and you do not need to install it as it is already installed.

- c) Power it on, username:kali, password: kali **[Attach the screenshot]**



Step 2: Install openssh-server on Splunk Forwarder Virtual machine.

We would download and **enable ssh on our Splunk Forwarder virtual machine** using the following commands

- a) First, we would run command to update our vm using `sudo apt update`
- b) Next, we run the command to install a ssh server on our ubuntu machine
`sudo apt install openssh-server`
- c) Run the command `sudo systemctl status ssh` to check the status of our ssh server. The status should be **green** showing running **[Attach the screenshot]**

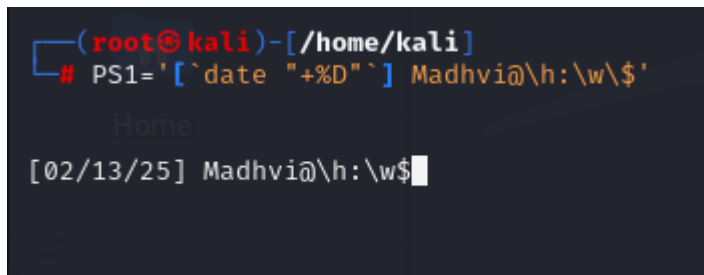
```
root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin# sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-02-13 08:35:22 EST; 5min ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 6769 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 6771 (sshd)
     Tasks: 1 (limit: 4552)
    Memory: 1.2M (peak: 1.5M)
       CPU: 10ms
    CGroup: /system.slice/ssh.service
            └─6771 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

- d) To make sure we enable ssh on ubuntu OS firewall just if the vm has one, execute the command `sudo ufw allow ssh`
- e) Check the IP address of your Ubuntu Universal Forwarder using `ifconfig`
- f) Use the command **who** to check and confirm your **username** on your Ubuntu Splunk forwarder.

Step 3: Login to Kali and Password Spray our Ubuntu Universal Forwarder machine (Victim Machine)- invalid Credentials (RED TEAM)

Let's have some fun. Let's play the attacker here.

- a) Login to your Kali VM using **username: kali** and **Password: Kali**. Use the command `PS1='[\`date "+%D"`] yourname@\h:\w\$ '` to set up your name on the terminal. **[Attach the screenshot]**

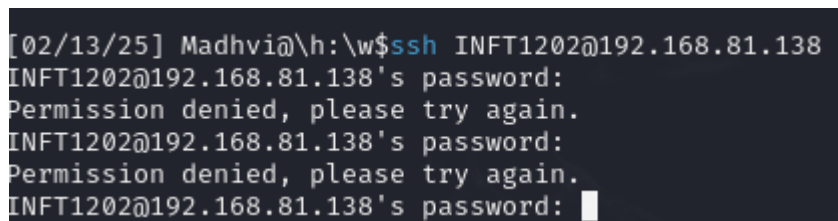


```
(root@kali)-[/home/kali]
# PS1='[\`date "+%D"`] Madhvi@\h:\w\$ '
Home
[02/13/25] Madhvi@\h:\w$
```

- b) Try to ssh with a wrong username as seen below using the **username INFT1202** and command would be

`ssh INFT1202@your_ubuntu_UniversalForwarderIP.`

[Attach the screenshot]



```
[02/13/25] Madhvi@\h:\w$ ssh INFT1202@192.168.81.138
INFT1202@192.168.81.138's password:
Permission denied, please try again.
INFT1202@192.168.81.138's password:
Permission denied, please try again.
INFT1202@192.168.81.138's password:
```

- c) As the Blue Team, on your Kali machine, run the command **ifconfig**. This is to confirm our Kali IP, as we would use this IP to check the logs in our first Ubuntu Splunk Server.

Lab 4: Detection of Attack by Splunk (Individual Lab)

INFT 1202 – Fuad Mustapha

Total Marks :40

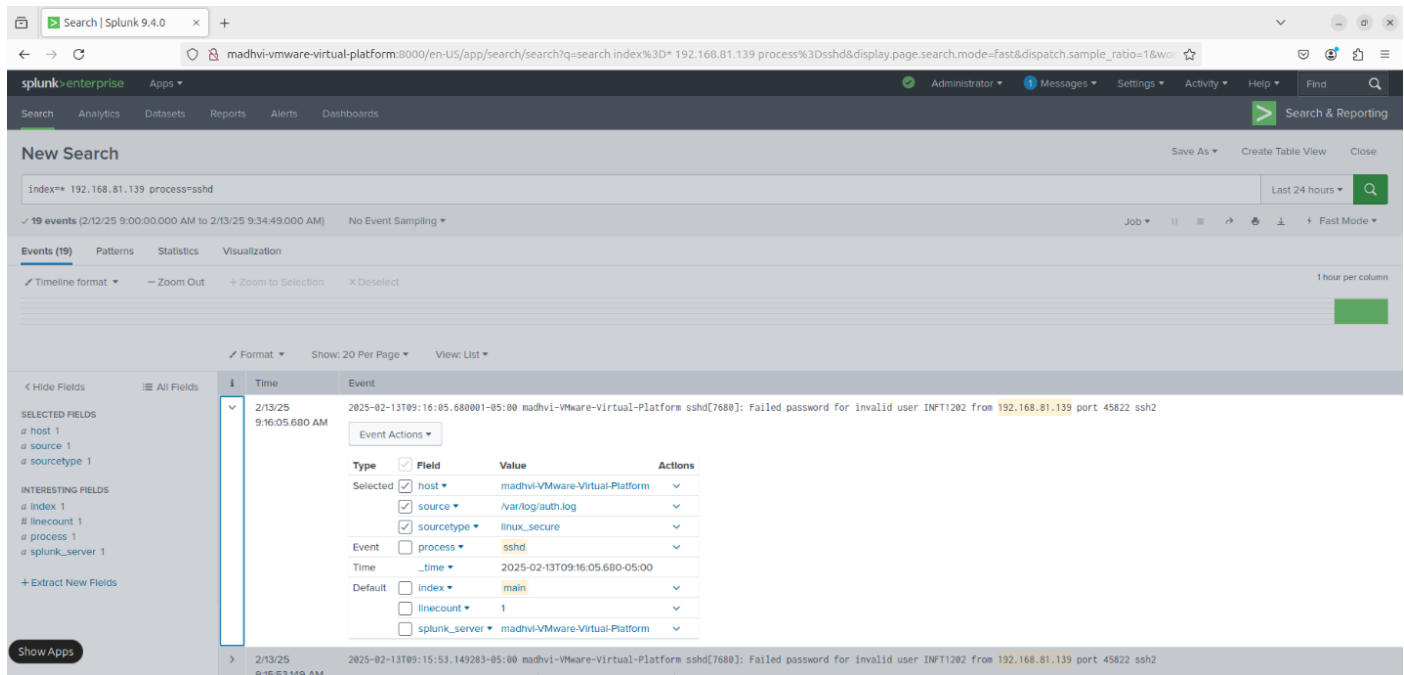
Step 4: Check the attack captured in first Ubuntu Splunk Server Machine.

Next, let us detect the attempted potential access to our Ubuntu machine using a wrong username.

Go to the Search and Reporting in Splunk and run the following command:

`index=* YourKaliIP` e.g., `index=* 192.168.182.135` (my Kali VM). We would

see that Splunk shows an invalid user is trying to access our Ubuntu virtual machine using ssh on port 22. **[Attach the screenshot]**



Lab 4: Detection of Attack by Splunk (Individual Lab)

INFT 1202 – Fuad Mustapha

Total Marks :40

The screenshot shows the Splunk Enterprise interface with a search query: `index=* 192.168.81.139 process=ssh`. The results are displayed in a table format, showing 19 events. The table has columns for Time and Event. The events are filtered by the search query and show various SSH login attempts, including failed password attempts and connection closures.

Time	Event
2025-02-13T09:16:05.680 AM	2025-02-13T09:16:05.680001-05:00 madhvi-VMware-Virtual-Platform sshd[7680]: Failed password for invalid user INFT1202 from 192.168.81.139 port 45822 ssh2
2025-02-13T09:15:53.149 AM	2025-02-13T09:15:53.149283-05:00 madhvi-VMware-Virtual-Platform sshd[7680]: Failed password for invalid user INFT1202 from 192.168.81.139 port 45822 ssh2
2025-02-13T09:15:50.954 AM	2025-02-13T09:15:50.954897-05:00 madhvi-VMware-Virtual-Platform sshd[7680]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.81.139
2025-02-13T09:15:38.343 AM	2025-02-13T09:15:38.343542-05:00 madhvi-VMware-Virtual-Platform sshd[7680]: Invalid user INFT1202 from 192.168.81.139 port 45822
2025-02-13T09:10:19.703 AM	2025-02-13T09:10:19.703555-05:00 madhvi-VMware-Virtual-Platform sshd[7518]: PAM 1 more authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.81.139
2025-02-13T09:10:19.703 AM	2025-02-13T09:10:19.703362-05:00 madhvi-VMware-Virtual-Platform sshd[7518]: Connection closed by invalid user Madhvi-NETW2200 192.168.81.139 port 32872 [preauth]

- a) You can see, everything got captured in Ubuntu Splunk Server from your kali machine in `/var/log/auth.log`

Good Job, now as the blue team member, we have detected an attacker is trying to log into our Ubuntu Universal Forwarder using an unidentified credential.

Step 5: Brute force attack on second Ubuntu Universal Forwarder machine by Kali machine by using Hydra tool in Kali Linux- Successful attack

- a) Go back to your Kali machine to play the attacker one more time and assume we do not know the password of the Ubuntu Universal Forwarder machine but know the username after doing some Open-Source Intelligence.
- b) Let's create a wordlist using random passwords **and add your Ubuntu Universal Forwarder VM password** to the list (see password list below), in this demo, my ubuntu password was very weak password which was

“Password”. To create password list, run the command on the terminal of your kali machine.

nano password.txt (This would open a **nano editor** named password.txt)

- c) Create a custom Password list by copying and pasting this password list to the opened editor and save it as password.txt.

Note: Make sure you add your ubuntu password to the list. This is to make our brute force faster.

```
PXyFaFexFxopg5k5
wfSTXaZyUTTfVrM9
YYh6nuqDJKgqSPtQ
E8XTUWQo9SMAJ7q6
GBLxJgQRgQiJ4DWC
iY1d1mcFM2KgZ7vr
MRfeALJUiUkAwWwp
nobCWwnNWcuKuzhz
9JmqVSq1V3jtJHxT
Password
Vagrant
```

- d) When the nano editor opens in your command line, copy the password list, and paste it to the editor and don't forget to include your correct ubuntu password to the editor (this would make our password spray faster)
- e) For **nano editor**, press ctrl + x and the editor would ask if you wanted to save. Type y(yes) to save our password list.
- f) Hope you guys are having some fun, there is still more fun playing the attacker here. Playing as the attacker and as stated earlier, we assume we already know the username of the Ubuntu machine so make sure you check the right username of your ubuntu machine by logging into your Ubuntu

Universal forwarder VM. Open the command terminal and execute the command - `who` **[Attach the screenshot]**

```
root@madhvi-VMware-Virtual-Platform:/opt/splunkforwarder/bin# who
madhvi    seat0      2025-02-13 08:29 (login screen)
madhvi    tty2       2025-02-13 08:29 (tty2)
madhvi    pts/1      2025-02-13 08:39
```

- g) After confirming the username of the Ubuntu Universal Forwarder VM, let's use **hydra** to password spray the ssh port of our Ubuntu machine using a brute force method and the custom password list. Execute the hydra command: `hydra -l [username] -P [password list] [victim machine IP] ssh` **[Attach the screenshot]**

```
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-02-13 10:27:15
[DATA] max 4 tasks per 1 server, overall 4 tasks, 4 login tries (l:1/p:4), ~1 try per task
[DATA] attacking ssh://192.168.81.138:22/
[ATTEMPT] target 192.168.81.138 - login "madhvi" - pass "madhvi@123" - 1 of 4 [child 0] (0/0)
[ATTEMPT] target 192.168.81.138 - login "madhvi" - pass "123" - 2 of 4 [child 1] (0/0)
[ATTEMPT] target 192.168.81.138 - login "madhvi" - pass "123@madhvi" - 3 of 4 [child 2] (0/0)
[ATTEMPT] target 192.168.81.138 - login "madhvi" - pass "madhvi" - 4 of 4 [child 3] (0/0)
[22][ssh] host: 192.168.81.138 login: madhvi password: 123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-02-13 10:27:19
```

Good Job, now we have gotten the victim's password (our Ubuntu Universal Forwarder VM) using the tool hydra to run a password spray.

- h) Playing the attacker, from our Kali VM, we would use the credentials to login to our Ubuntu Universal Forwarder VM through SSH. **[Attach the screenshot]**

```
# ssh YourUsername@[UbuntuServerVMIP]
```

```
# ssh fuadnetw2200@192.168.80.140
```

```
[02/13/25] Madhvi@h:\w$ssh madhvi@192.168.81.138
madhvi@192.168.81.138's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.11.0-17-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

219 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Good Job, playing the attacker, you have successfully gained access to the Ubuntu Universal Forwarder VM.

Step 6: Check the successful attack logs in your first Ubuntu Splunk server machine

- a) Check the successful attempt logs captured by Ubuntu Splunk Server **[Attach the screenshot]**

Time

Event

2/13/25
10:31:12.655 AM

2025-02-13T10:31:12.655341-05:00 madhvi-VMware-Virtual-Platform sshd[8564]: Accepted password for madhvi from 192.168.81.139 port 46530 ssh2

Event Actions

Type	☒ Field	Value	Actions
Selected	☒ host	madhvi-VMware-Virtual-Platform	
	☒ source	/var/log/auth.log	
	☒ sourcetype	linux_secure	
Time	☐ _time	2025-02-13T10:31:12.655-05:00	
Default	☐ index	main	
	☐ linecount	1	
	☐ splunk_server	madhvi-VMware-Virtual-Platform	

- b) Screenshot confirming our Ubuntu username and we would see Splunk detected the successful login from the attacker machine

Step 7: Answer few questions based on the scenario we saw.

Good Job, now we have successfully completed this lab. From the screenshot for our detection, we need some important details to use for our investigation as Security Analysts.

Answer the following questions based on your screenshots: (below are my answers, you should get yours)

- a) What is the attacker IP detected by Splunk: **192.168.81.139**
- b) Based on your screenshot, what port did Splunk detect: **46530**
- c) What port was attacked (default ssh port): **8564**
- d) What is the source type: linux_secure
- e) What is the log source: **/var/log/auth_log**

Congratulations on finishing your investigation.

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 15 Screenshots each 1 mark = 30 marks
 - b) Video Demonstration = 5 marks
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professor's from lab manual. They are for your reference only and would be considered an academic integrity breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**
- **Submit a video recording (sharing screen) briefly explaining the steps taken to perform the lab and show each of the VM running. Execute the commands in your VMs to show the domain and network configuration. If the video recording stating the above requirements are not submitted, you will be eligible to get marks out of 60% of Lab 4 marks.**